

Lead & Tender Detection Engine

Phase 1 — Multi-source tender scraping, cleaning and lead scoring

CSTAM 3.0 Challenge — OliveSoft Automated RFP Intelligence System

Component	Public tender discovery, parsing, cleaning, relevance classification and lead scoring
Sources	HAICOP / marchespublics.gov.tn (rich detail pages) and appeloffres.com (public listing)
Main file	cstam_pipeline.py (single file, also runs as one Colab cell)
Outputs	offers_combined.json, offers_combined.csv, offers_report.pdf, offers_quality_report.json
Version	5 — internal engineering documentation

1 Overview

This module implements Phase 1 of the challenge. It crawls two public procurement portals, extracts one structured record per tender, removes noise (expired tenders, non-IT tenders, duplicates), scores each remaining tender for OliveSoft, and exports the leads in formats meant for the next phases: a JSON file for the RAG and agent stages, a CSV for the dashboard, a PDF in the same layout as the sample RFP, and an optional webhook call to n8n.

It is self-contained in one Python file. Nothing is invented: a field the source does not publish stays empty and is listed in `missing_fields`. No mock data is used unless `--demo` is passed, in which case every record is labelled `simulated`.

2 Architecture

2.1 Data sources

- **HAICOP** — official portal. The listing is rendered by JavaScript, so tender ids are collected with Playwright (headless Chromium); each detail page is then fetched with `requests`. Detail pages give buyer, object, description, lots, deadline with hour, evaluation method and links to public documents (PDF, DOCX, DOC), which are read in memory only.
- **appeloffres.com** — public, server-rendered category listings (Informatique, Telecom). Only the listing is read: buyer code, truncated title, reference, dates and URL. Detail pages, buyer names and specifications sit behind a login and are not accessed.
- Both crawlers respect `robots.txt`, wait 1.5–3.5 s between requests and retry up to 3 times with back-off. If one source fails or is blocked, it is reported and skipped; the other still runs.

2.2 Pipeline

Discovery → fetch and parse → cross-portal linking → cleaning and enrichment → relevance and lead scoring → filter (expired, below fit) → export (JSON, CSV, PDF, quality report, optional webhook).

- **Cross-portal linking**: an appeloffres.com row with the same deadline date and a title similarity of at least 0.8 (accent-insensitive, first 70 characters, because titles are truncated) is folded into the matching HAICOP record, which keeps the richer data and lists the other portal in `also_listed_on`.
- **Document reading**: to save requests, HAICOP documents are downloaded only when the page text already looks IT (IT score of at least 1, or a Type de commande containing “informatique”). At most 3 documents per tender.
- **New-lead tracking**: `is_new` and `first_seen_at` are computed against the previous `offers_combined.json` (matched on URL).

3 Relevance and Lead Scoring

Relevance is decided in two steps, then a lead score is computed for the sales dashboard.

3.1 IT keyword score (first filter)

Weighted, accent-insensitive whole-word matching on object, description, lots and document text: strong terms (ERP, cloud, logiciel, cybersecurite...) count 2 points, weak terms (reseau, application, internet...) 1 point. Known false friends are removed before scoring (reseau d'eau, developpement agricole, stockage d'energie, batteries, hebergement of apprentices...). A tender is kept when its score reaches 2 (HAICOP) or 1 (appeloffres.com, whose categories are already IT), or when HAICOP's own Type de commande is informatique.

3.2 Segment and OliveSoft fit (second filter)

Segment	Fit	Segment	Fit
AI / data / BI	high	Network / telecom	low
Software development / apps / ERP	high	Licences / off-the-shelf software	low
Cybersecurity	medium	Hardware supply	low
Cloud / servers / infrastructure	medium	Generic IT	low
CCTV / IoT, consumables, digital marketing, other: fit <i>none</i>			

Office supplies (toners, cartridges) are recognised as consumables, not IT projects. The gate `--min-fit` (default *low*) drops anything below the chosen level; dropped rows are always logged and written to the quality report, never silently removed.

3.3 Lead score (0–100)

Component	Points
Fit: high / medium / low / none	60 / 40 / 20 / 0
IT keyword score, capped at 10	1.5 per point (max 15)
Data completeness (0–100)	completeness divided by 10 (max 10)
Deadline within 7 days (urgent) / within 14 days	+10 / +5
Expired tender (only with <code>--keep-expired</code>)	score forced to 0

Priority label: **hot** at 70 or more, **warm** at 45 or more, otherwise **cold**. Completeness weights: buyer 25, object 10, untruncated title 5, description / lots / document text 20, deadline 15, deadline hour 5, documents 10, extracted sections 10.

4 Output Data Model

One JSON object per tender, in a JSON array (`offers_combined.json`), rewritten on each run. The CSV holds the same columns; lists and objects are JSON-encoded inside cells. Main fields:

Field	Description
numero_ao, tender_id, notice_kind	Reference (placeholders replaced by the id), unique id, kind: appel_offres / consultation / manifestation_interet / report
acheteur, buyer_hint	Buyer as published; hint from the title (university, hospital, municipality...) when absent
objet, objet_ar, descriptif	French title (Arabic-only titles are kept in objet_ar and replaced by the French description), full description
segment, olivesoft_fit, lead_score, priority	Classification and dashboard ranking (section 3)
it_score, it_terms, skills_detected	Keyword score, matched terms, technologies found (whole-word match)
lots[]	Parsed lots: number, object, quantity, region, provisional guarantee (one block per lot)
requirements[]	Requirement lines with level exige / souhaite / non-precise, taken only from source text
date_limite, heure_limite, deadline_at, days_left, urgent, expired	Deadline in Tunisia time (UTC+1). "00:00" means the hour was not published
documents[], document_sections, doc_excerpt	Public document links, extracted sections (prescriptions, personnel, references, criteria), first 4000 characters
rag_query	Ready-made semantic search query for the RAG stage (object, description, lots, domain, keywords)
completeness, detail_level, quality_flags, missing_fields	Data-quality signals (section 5)
also_listed_on, source_site, url, is_new, first_seen_at	Provenance and new-lead tracking

Other outputs

- **offers_report.pdf** — one tender per page in the layout of **sample_rfp_fr.pdf** (header, Client, Secteur, Date limite, Budget previsionnel, numbered technical prescriptions, personnel, references, criteria). Empty sections are omitted. Text is ASCII: Arabic is dropped, accents are removed.
- **offers_quality_report.json** — counts by source, fit, priority, segment, detail level, missing fields and flags, plus every dropped tender with its reason.

5 Edge Cases and Data Validation

Situation	Handling
Source blocked, layout changed, login redirect	Source reported and skipped, debug HTML saved, other source continues
Missing buyer	<code>buyer_unknown</code> flag, <code>buyer_hint</code> from the title
Placeholder reference (“...”, “Consultation”)	Replaced by the tender id, <code>reference_missing</code> flag, notice kind extracted
Truncated title (“...”) Arabic-only title	<code>title_truncated</code> flag, lower completeness Kept in <code>objet_ar</code> , French description used as title (<code>title_arabic_only</code> flag if none)
Deadline hour not published	<code>deadline_time_unknown</code> flag, day treated as running until 23:59
Expired tender	Dropped, hour-aware, in Tunisia time (kept with <code>--keep-expired</code>)
Postponement notice	<code>postponement_notice</code> flag
No requirement text at all	<code>no_requirements_text</code> flag, <code>listing_only</code> detail level
Site footer text mixed into lots	Cut before parsing
False-friend IT words, office supplies	Excluded before scoring; consumables get fit <i>none</i>
Unreadable or oversized document	Skipped (limit 15 MB); <code>.doc</code> files need <code>antiword</code> installed
Nothing collected	Valid empty JSON array and a PDF stating that nothing was collected

6 Usage and Configuration

Colab: paste the whole file in one cell and run it; options go in the `RUN_ARGS` string at the top. Local machine:

```
pip install -r requirements.txt
playwright install chromium
python cstam_pipeline.py --min-fit medium
```

Option	Effect
<code>--sources haicop,appeloffres</code>	Sources to run (default both)
<code>--min-fit none low medium high</code>	OliveSoft relevance gate (default low)
<code>--enrich</code>	Scrape HAICOP without its IT filter so appeloffres.com rows can be linked to a HAICOP record
<code>--hc-pages, --max-tenders, --pages</code>	HAICOP listing pages (8), HAICOP detail pages per run (80), appeloffres.com pages per category (5)
<code>--keywords "sig,gps"</code>	Extra IT keywords (2 points each)
<code>--ids Tender-98007,...</code>	Skip the HAICOP listing and use ids copied from the browser
<code>--keep-expired, --all</code>	Keep expired tenders, disable all IT filters
<code>--webhook URL</code>	POST new medium/high leads as <code>\{count, leads[]\}</code> to an n8n Webhook node
<code>--demo</code>	Simulated, labelled tenders that exercise every edge case; no network
<code>--browser-ua, --headed, --out-dir, --prefix</code>	Browser User-Agent, visible browser, output folder, file name prefix

7 Known Limitations & Next Steps

- **Multi-lot tenders.** The previous version captured only the first lot’s region and budget from a whole-page label search. This version parses each `Lot N` block as its own structure (`lots[]`, with region and provisional guarantee per lot). It has been checked on a synthetic page only and still needs verifying on a live multi-lot tender such as Tender-103899.
- **TUNEPS** is intentionally excluded (`robots.txt` disallow, prior HTTP 403 responses); revisit only via an official API or partnership channel.
- **WebManagerCenter** (private and parapublic tender news) is not part of this version. Its article structure

is theme-dependent and its search index mixes recent notices with decade-old archives, so it needs a freshness filter and closer monitoring before being re-added as a third source.

- **Scoring is keyword- and category-based.** A natural next step (Phase 2 of the challenge) is to route qualifying tenders into the RAG layer for deeper semantic matching against OliveSoft's project history.
- **No scheduling or orchestration** is included here by design: this module is meant to be invoked by the n8n workflow layer.
- **Portal access.** HAICOP may filter cloud IP addresses (Colab). If the listing comes back empty, run locally from a Tunisian connection or use `--ids`. Listing selectors depend on the current page layout and may need maintenance.
- **Terms of use.** `appeloffres.com` is a private aggregator whose conditions may restrict reuse of its content. Only its public listing is read and no login is bypassed; check its terms before any use beyond the demo, or run with `--sources haicop`.
- **Live verification.** Cleaning, scoring, export and the parsers were tested on simulated and synthetic input; the first real crawl should be checked against the portals' current pages.